1) a)

$\{x: x \in \mathbb{R}, x \neq 1\}$

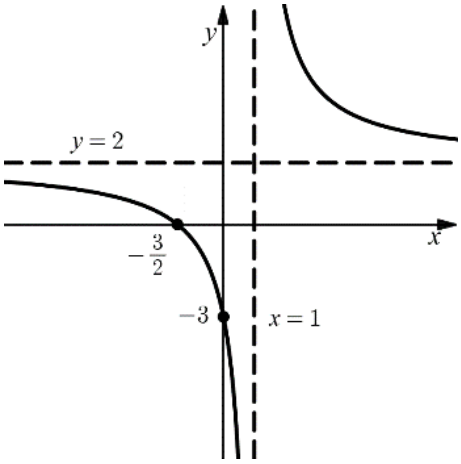
1) b)

$x = 1$ and $y = 2$

1) c)

$\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \neq 2\}$

1) d)



2) a)

$\left\{x: x \in \mathbb{R}, x \neq \frac{4}{3}\right\}$

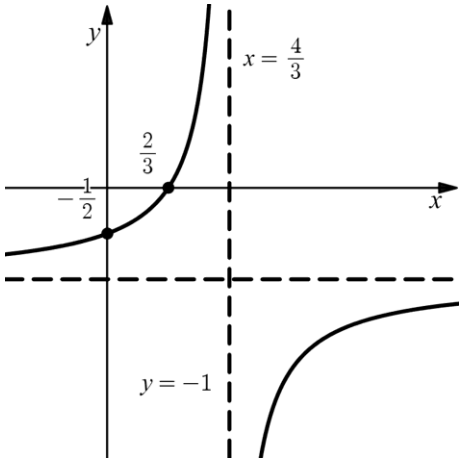
2) b)

$x = \frac{4}{3}$ and $y = -1$

2) c)

$\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \neq -1\}$

2) d)



3) a)

$x \in \mathbb{R}$

3) b)

$y = 1$

3) c)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-1}{x^2+4} \Rightarrow k(x^2 + 4) = x^2 - 1$

$(k - 1)x^2 + (4k + 1) = 0$ for $k \neq 1$

For k to exist: $0 - 4(k - 1)(4k + 1) \geq 0$

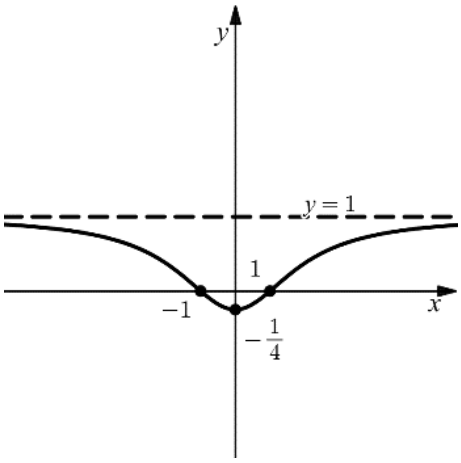
$(k - 1)(4k + 1) \leq 0$

$\therefore -\frac{1}{4} \leq k \leq 1$

When $k = 1$, there is an asymptote $\therefore -\frac{1}{4} \leq k < 1$

$\therefore \left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, -\frac{1}{4} \leq \mathrm{f}(x) < 1\right\}$

3) d)



4) a)

$x^2 - x - 2 = (x - 2)(x + 1)$

$\therefore x = -1, x = 2$ and $y = 3$

4) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{3x^2 - x - 2}{x^2 - x - 2} \Rightarrow k(x^2 - x - 2) = 3x^2 - x - 2$

$(k - 3)x^2 + (1 - k)x + (2 - 2k) = 0$ for $k \neq 3$

For k to exist: $(1 - k)^2 - 4(k - 3)(2 - 2k) \geq 0$

$9k^2 - 34k + 25 \geq 0$

$(k - 1)(9k - 25) \geq 0$

$\therefore k \leq 1$ or $k \geq \frac{25}{9}$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 1\} \cup \left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq \frac{25}{9}\right\}$

4) c)

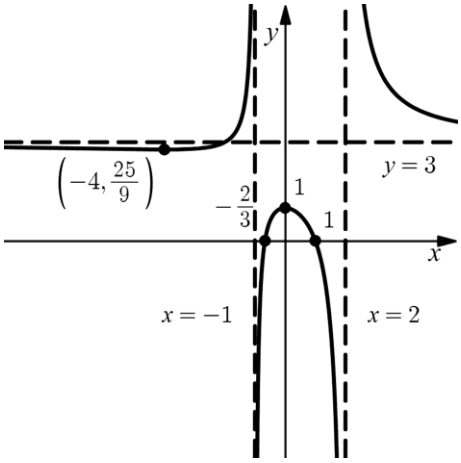
When $y = 1$, $1 = \frac{3x^2 - x - 2}{x^2 - x - 2} \Rightarrow x^2 - x - 2 = 3x^2 - x - 2 \Rightarrow 2x^2 = 0 \therefore x = 0$

When $y = \frac{25}{9}$, $\frac{25}{9} = \frac{3x^2 - x - 2}{x^2 - x - 2} \Rightarrow 25(x^2 - x - 2) = 9(3x^2 - x - 2) \Rightarrow 2x^2 + 16x + 32 = 0$

$\therefore x = -4$

Local min at and local max at $(0, 1)$

4) d)



5) a)

$\{x: x \in \mathbb{R}, x \neq -2\}$

5) b)

$x = -2, y = 1$

5) c)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-1}{(x+2)^2} \Rightarrow k(x+2)^2 = x^2 - 1$

$(k-1)x^2 + 4kx + 4k + 1 = 0$ for $k \neq 1$

For k to exist: $(4k)^2 - 4(k-1)(4k+1) \geq 0$

$12k + 4 \geq 0$

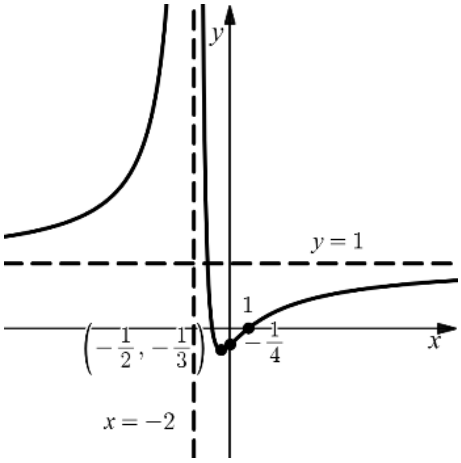
$\therefore k \geq -\frac{1}{3}$

So there is a minimum when $y = -\frac{1}{3}$

5) d)

$\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq -\frac{1}{3}\right\}$

5) e)



6) a)

$\{x: x \in \mathbb{R}, x \neq -5\} \cup \{x: x \in \mathbb{R}, x \neq -1\}$

6) b)

$x = -5,\ x = -1,\ y = a$

6) c)

$a = 3$

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{3x^2+6}{x^2+6x+5} \Rightarrow k(x^2 + 6x + 5) = 3x^2 + 6$

$(k-3)x^2 + 6kx + 5k - 6 = 0$ for $k \neq 3$

For k to exist: $(6k)^2 - 4(k-3)(5k-6) \geq 0$

$16k^2 + 84k - 72 \geq 0$

$\therefore k \geq \frac{3}{4}$ or $k \leq -6$

$\therefore y = -6$ and $y = \frac{3}{4}$

6) d)

$\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -6\} \cup \left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq \frac{3}{4}\right\}$

7) a)

$\{x: x \in \mathbb{R}\}$

7) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{2x^2+3}{x^2+1} \Rightarrow k(x^2+1) = 2x^2+3$

$(2-k)x^2 + 3 - k = 0$ for $k \neq 2$

For k to exist: $0^2 - 4(2-k)(3-k) \geq 0$

$-4k^2 + 20k - 24 \geq 0$

$k^2 - 5k + 6 \leq 0$

$\therefore 2 < k \leq 3$ as $k \neq 2$

$\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, 2 < \mathrm{f}(x) \leq 3\}$

7) c)

When $y = \frac{5}{2}$, $\frac{5}{2} = \frac{2x^2+3}{x^2+1} \Rightarrow 5(x^2+1) = 2(2x^2+3) \Rightarrow x^2 = 1 \therefore x = \pm 1$

So $-1 \leq x \leq 1$

$\therefore \{x: x \in \mathbb{R}, -1 \leq x \leq 1\}$

8) a)

$\{x: x \in \mathbb{R}, x \neq -2\} \cup \{x: x \in \mathbb{R}, x \neq 2\}$

8) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-6x+16}{x^2-4} \Rightarrow k(x^2-4) = x^2 - 6x + 16$

$(1-k)x^2 - 6x + 16 + 4k = 0$ for $k \neq 1$

For k to exist: $(-6)^2 - 4(1-k)(16+4k) \geq 0$

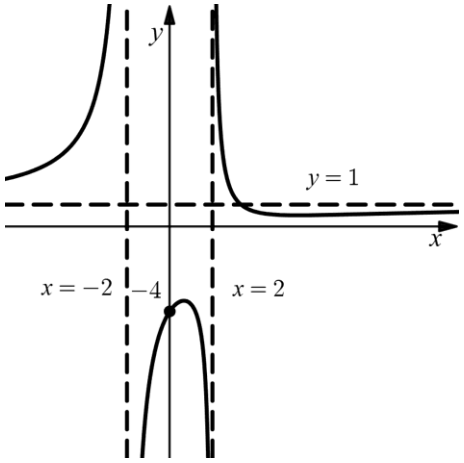
$16k^2 + 48k - 28 \geq 0$

$4k^2 + 9k - 4 \geq 0$

$\therefore k \geq \frac{1}{2}$ or $k \leq -\frac{7}{2}$

$\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -\frac{7}{2}\right\} \cup \left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq \frac{1}{2}\right\}$

8) c)



8) d)

When $y = -4,\ -4 = \frac{x^2-6x+16}{x^2-4}$

$\Rightarrow -4(x^2-4) = x^2 - 6x + 16 \Rightarrow 5x^2 - 6x = 0 \therefore x = 0$ or $x = \frac{6}{5}$

When $y = 1,\ 1 = \frac{x^2-6x+16}{x^2-4}$

$\Rightarrow x^2 - 4 = x^2 - 6x + 16 \Rightarrow 6x = 20 \therefore x = \frac{10}{3}$

$\therefore 0 \leq x \leq \frac{6}{5}$ or $x > \frac{10}{3}$

$\left\{x: x \in \mathbb{R}, 0 \leq x \leq \frac{6}{5}\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{10}{3}\right\}$

9) a)

$x^2 + ax + 4 \geq 0$

Consider the discriminant: $b^2 - 4ac \leq 0$

$a^2 - 4 \times 4 \leq 0$

$\therefore a^2 \leq 16$ as req.

9) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2+4x+4}{x^2+1} \Rightarrow k(x^2+1) = x^2+4x+4$

$(1-k)x^2 + 4x + 4 - k = 0$ for $k \neq 1$

For k to exist: $4^2 - 4(1-k)(4-k) \geq 0$

$-4k^2 + 20k \geq 0$

$\therefore 0 \leq k \leq 5$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, 0 \leq \mathrm{f}(x) \leq 5\}$

9) c)

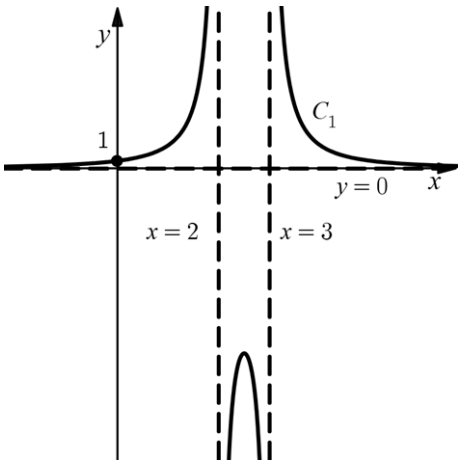
When $y = 2$, $2 = \frac{x^2+4x+4}{x^2+1}$

$\Rightarrow 2(x^2+1) = x^2+4x+4 \Rightarrow x^2 - 4x - 2 = 0 \therefore x = 2-\sqrt{6}$ or $x = 2+\sqrt{6}$

$\therefore x \leq 2-\sqrt{6}$ or $x \geq 2+\sqrt{6}$

$\{x: x \in \mathbb{R}, x \leq 2-\sqrt{6}\} \cup \{x: x \in \mathbb{R}, x \geq 2+\sqrt{6}\}$

10) a)



10) b)

$\left(\frac{6}{x^2-5x+6}\right)^2 = \frac{6}{x^2-5x+6} \Rightarrow 36(x^2-5x+6) = 6(x^2-5x+6)^2$

Since $x^2 - 5x + 6 \neq 0$ for the curve to exist:

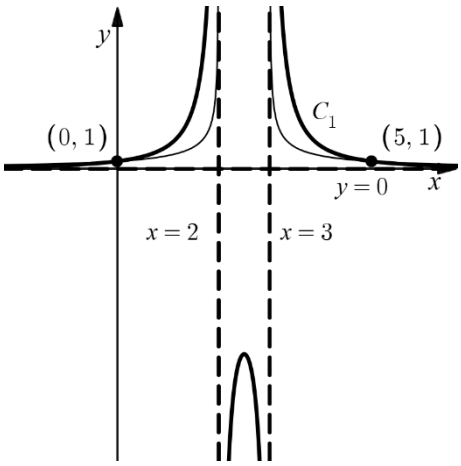
$6 = x^2 - 5x + 6 \Rightarrow x^2 - 5x = 0 \Rightarrow x(x-5) = 0 \therefore x = 0, x = 5$

When $x = 0$, $y = 1$, when $x = 5$, $y = 1$

$\therefore (0,1)$ and $(5,1)$

10) c)

The curves $y = \frac{6}{x^2-5x+6}$ and $y = \sqrt{\frac{6}{x^2-5x+6}}$ intersect at $x = 0$ and $x = 5$



$0 < x < 2$ or $3 < x < 5$

$\therefore \{x: x \in \mathbb{R}, 0 < x < 2\} \cup \{x: x \in \mathbb{R}, 3 < x < 5\}$

11) a)

$\frac{x^2+1}{x} = x + \frac{1}{x}$

Asymptotes: $x = 0$ and $y = x$

$\therefore x = 0$ and $y = x$

11) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2+1}{x} \Rightarrow kx = x^2 + 1$

$x^2 - kx + 1 = 0$

For k to exist: $(-k)^2 - 4(1)(1) \geq 0$

$k^2 - 4 \geq 0$

$\therefore k \geq 2$ or $k \leq -2$

So turning points are when $y = -2$ and $y = 2$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -2\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 2\}$

11) c)

Since $y = x$ is the oblique asymptote, then $x > 0$

$\{x: x \in \mathbb{R}, x > 0\}$

12) a)

	x	-2	
x	x^2	$-2x$	$+1$
$+2$	$+2x$	-4	

$\frac{x^2-3}{x+2} = x - 2 + \frac{1}{x+2}$

Asymptotes: $x = -2$ and $y = x - 2$

$\therefore x = -2$ and $y = x - 2$

12) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-3}{x+2} \Rightarrow k(x+2) = x^2 - 3$

$x^2 - kx - 3 - 2k = 0$

For k to exist: $(-k)^2 - 4(1)(-3-2k) \geq 0$

$k^2 + 8k + 12 \geq 0$

$\therefore k \geq -2$ or $k \leq -6$

So turning points are when $y = -6$ and $y = -2$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -6\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq -2\}$

12) c)

Since $y = x - 2$ is the oblique asymptote, then $x < -2$

$\{x: x \in \mathbb{R}, x < -2\}$

13) a)

$\{x: x \in \mathbb{R}, x \neq 2\}$

13) b)

	$3x$	$+4$	
x	$3x^2$	$+4x$	$+27$
-2	$-6x$	-8	

$\frac{3x^2-2x+19}{x-2} = 3x + 4 + \frac{27}{x-2}$

Asymptotes: $x = 2$ and $y = 3x + 4$

13) c)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{3x^2-2x+19}{x-2} \Rightarrow k(x-2) = 3x^2 - 2x + 19$

$3x^2 + (-2-k)x + 19 + 2k = 0$

For k to exist: $(-2-k)^2 - 4(3)(19+2k) \geq 0$

$k^2 - 20k - 224 \geq 0$

$\therefore k \geq 28$ or $k \leq -8$

So turning points are when $y = -8$ and $y = 28$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -8\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 28\}$

14) a)

$\{x: x \in \mathbb{R}, x \neq -1\}$

14) b)

	$2x$	$+3$	
x	$2x^2$	$+3x$	-6
$+1$	$+2x$	$+3$	

$\frac{2x^2+5x-3}{x+1} = 2x + 3 - \frac{6}{x+1}$

Asymptotes: $x = -1$ and $y = 2x + 3$

14) c)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{2x^2+5x-3}{x+1} \Rightarrow k(x+1) = 2x^2 + 5x - 3$

$2x^2 + (5-k)x - 3 - k = 0$

For k to exist: $(5-k)^2 - 4(2)(-3-k) \geq 0$

$k^2 - 2k + 49 \geq 0$

$(k-1)^2 + 48 \geq 0$

Since $(k-1)^2 \geq 0$ and $48 > 0$ this is true for all k

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}\}$

15) a)

$\{x: x \in \mathbb{R}, x \neq -4\}$

15) b)

	x	$+(a-4)$	
x	x^2	$(a-4)x$	$17-4a$
$+4$	$+4x$	$4a-16$	

$\frac{x^2+ax+1}{x+4} = x+a-4+\frac{17-4a}{x+1}$

Asymptotes: $x=-1$ and $y=x+a-4$

$\therefore a=4$

15) c)

Find the values for which $y=k$ intersects $y=\mathrm{f}(x)$:

$k=\frac{x^2+4x+1}{x+4} \Rightarrow k(x+4)=x^2+4x+1$

$x^2+(4-k)x+1-4k=0$

For k to exist: $(4-k)^2-4(1-4k)\geq 0$

$k^2+8k+12\geq 0$

$\therefore k\geq -2$ or $k\leq -6$

So turning points are when $y=-6$ and $y=-2$

$\therefore \{\mathrm{f}(x):\mathrm{f}(x)\in\mathbb{R},\mathrm{f}(x)\leq -6\}\cup\{\mathrm{f}(x):\mathrm{f}(x)\in\mathbb{R},\mathrm{f}(x)\geq -2\}$

16) a)

	x	$3+a$	
x	x^2	$(3+a)x$	$15+3a$
-3	$-3x$	$-9-3a$	

$\frac{x^2+ax+4}{x-3} = x+3+a+\frac{15+3a}{x-3}$

Asymptotes: $x=3$ and $y=x+3+a$

$\therefore y=x+3+a$

16) b)

$(0,-1)$ gives $-1=0+3+a$

$\therefore a=-4$

16) c)

$\{x: x\in\mathbb{R}, x\neq 3\}$

16) d)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-4x+4}{x-3} \Rightarrow k(x-3) = x^2 - 4x + 4$

$x^2 + (-4-k)x + 4 + 3k = 0$

For k to exist: $(-4-k)^2 - 4(1)(4+3k) \geq 0$

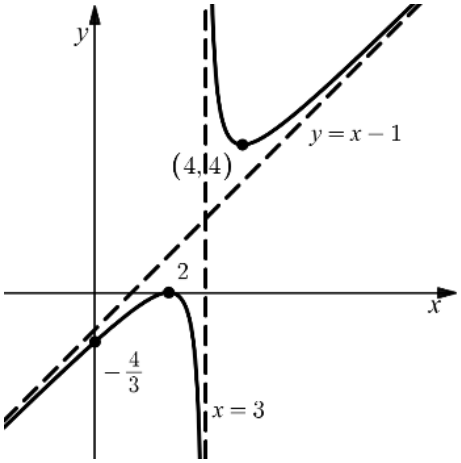
$k^2 - 4k \geq 0$

$\therefore k \geq 4$ or $k \leq 0$

So turning points are when $y = 0$ and $y = 4$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 0\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 4\}$

16) e)



17) a)

	$2x$	$+5$	
x	$2x^2$	$+5x$	$+4$
-2	$-4x$	-10	

$\frac{2x^2+x-6}{x-2} = 2x + 5 + \frac{4}{x-2}$

Asymptotes: $x = 2$ and $y = 2x + 5$

$\therefore x = 2$ and $y = 2x + 5$

17) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{2x^2+x-6}{x-2} \Rightarrow k(x-2) = 2x^2 + x - 6$

$2x^2 + (1-k)x + 2k - 6 = 0$

For k to exist: $(1-k)^2 - 4(2)(2k-6) \geq 0$

$k^2 - 18k + 49 \geq 0$

$\therefore k \geq 9 + 4\sqrt{2}$ or $k \leq 9 - 4\sqrt{2}$

So turning points are when $y = 9 - 4\sqrt{2}$ and $y = 9 + 4\sqrt{2}$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 9 - 4\sqrt{2}\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 9 + 4\sqrt{2}\}$

17) c)

Since $y = 2x + 5$ is the oblique asymptote, we know $x > 2$

Consider when $y = 2x + 6$ intersects the curve:

$2x + 6 = \frac{2x^2+x-6}{x-2} \Rightarrow (2x+6)(x-2) = 2x^2 + x - 6 \Rightarrow 2x^2 + 2x - 12 = 2x^2 + x - 6$

$\Rightarrow x = 6$

So $2 < x < 6$

$\therefore \{x: x \in \mathbb{R}, 2 < x < 6\}$

18) a)

	x	$a-1$	
x	x^2	$(a-1)x$	$-a$
$+1$	$+x$	$a-1$	

$\frac{x^2+ax-1}{x+1} = x + a - 1 - \frac{a}{x+1}$

Asymptotes: $x = -1$ and $y = x + a - 1$

$\therefore x = -1$ and $y = x + a - 1$

18) b)

$\frac{1}{3} = \frac{2^2+2a-1}{2+1} \Rightarrow 4 + 2a - 1 = 2 \therefore a = -1$

18) c)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-x-1}{x+1} \Rightarrow k(x+1) = x^2 - x - 1$

$x^2 + (-1-k)x + (-k-1) = 0$

For k to exist: $(-1-k)^2 - 4(1)(-k-1) \geq 0$

$k^2 + 6k + 5 \geq 0$

$\therefore k \leq -5$ or $k \geq -1$

So turning points are when $y = -5$ and $y = -1$

$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -5\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq -1\}$

18) d)

Since $y = x - 2$ is the oblique asymptote we know $x > -1$

Consider when $y = x + 1$ intersects the curve:

$x + 1 = \frac{x^2-x-1}{x+1} \Rightarrow (x+1)(x+1) = x^2 - x - 1 \Rightarrow x^2 + 2x + 1 = x^2 - x - 1$

$\Rightarrow x = -\frac{2}{3}$

So $x \leq -\frac{2}{3}$ or $x > -1$

$\therefore \left\{x: x \in \mathbb{R}, -1 < x \leq -\frac{2}{3}\right\}$

19) a)

	x	-1	
x	x^2	$-x$	$+4$
-1	$-x$	$+1$	

$\frac{x^2-2x+5}{x-1} = x - 1 + \frac{4}{x-1}$

Asymptotes: $x = 1$ and $y = x - 1$

$\therefore x = 1$ and $y = x - 1$

19) b)

Find the values for which $y = k$ intersects $y = \mathrm{f}(x)$:

$k = \frac{x^2-2x+5}{x-1} \Rightarrow k(x-1) = x^2 - 2x + 5$

$x^2 + (-2-k)x + k + 5 = 0$

For k to exist: $(-2-k)^2 - 4(k+5) \geq 0$

$k^2 - 16 \geq 0$

$(k-4)(k+4) \geq 0$

$\therefore k \geq 4$ or $k \leq -4$

So turning points are when $y = 4$ and $y = -4$

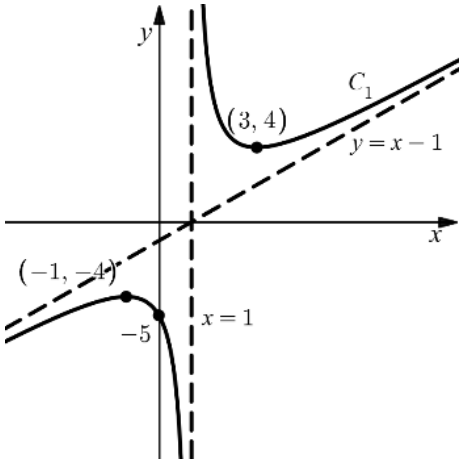
$\therefore \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq -4\} \cup \{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 4\}$

19) c)

When $y = -4, x = -1$

When $y = 4$, $x = 3$

When $x = 0$, $y = -5$



19) d)

Since $\mathrm{g}(x) = [\mathrm{f}(x)]^2$

For intersection $\mathrm{f}(x) = \mathrm{g}(x)$ and so $\mathrm{f}(x) = [\mathrm{f}(x)]^2$

$\Rightarrow \mathrm{f}(x)\big(1 - \mathrm{f}(x)\big)=0$

So either $\mathrm{f}(x) = 0 \Rightarrow \frac{x^2-2x+5}{x-1} = 0$ but $x^2 - 2x + 5$ has no real roots since $4 - 20 < 0$

Or $\mathrm{f}(x) = 1 \Rightarrow \frac{x^2-2x+5}{x-1} = 1 \Rightarrow x^2 - 2x + 5 = x - 1$

$\Rightarrow x^2 - 3x + 6 = 0$ which has no real roots since $9 - 24 < 0$

So there are no points of intersection.

19) e)

$\frac{x^2-2x+5}{x-1} > \sqrt{\frac{x^2-2x+5}{x-1}}$

$f(x) > \sqrt{f(x)}$

Consider $f(x) \geq 0$ only, due to the square root

Square both sides:

$[f(x)]^2 > f(x)$

In part d) we saw that there were no points where $f(x) = [f(x)]^2$

So that $[f(x)]^2 > f(x)$ for $f(x) \geq 4$

$f(x) \geq 4$ when $x > 1$

$\therefore \{x: x \in \mathbb{R}, x > 1\}$